

SEARCHING

Q9)

Question:

DS	Session	Searching	Question Information	Level 1	Challenge 9
Problem Description: Prabhu Salamon is planning to make a very long journey across the cityside by Train. His journey consists of N train routes, numbered from 1 to N in the order he must take them. The trains themselves are very fast, but do not run often. The i-th train route only runs every X_i days. More specifically, he can only take the i-th train on day $X_i, 2X_i, 3X_i$ and so on. Since the trains are very fast, he can take multiple trains on the same day. Prabhu Salamon must finish his journey by day D, but he would like to start the journey as late as possible. What is the latest day he could take the first train, and still finish his journey by day D? It is guaranteed that it is possible for Prabhu Salamon to finish his journey by day D. Constraints: $1 \leq T \leq 100$. $1 \leq X_i \leq D$. $1 \leq N \leq 1000$. $1 \leq D \leq 10^{12}$ Input Format: The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with a line containing the two integers N and D. Then, another line follows containing N integers, the i-th one is X_i. Output Format: Print the output in a single line contains, the latest day he could take the first train, and still finish his journey by day D.					

PROGRAM:

```
1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      int t, n, i;
6      double d, x;
7
8      printf("Input:\n");
9
10     scanf("%d", &t);
11
12     while (t--) {
13         scanf("%d %lf", &n, &d);
14
15         double a[1000];
16
17         for (i = 0; i < n; i++)
18             scanf("%lf", &a[i]);
19
20         for (i = n - 1; i >= 0; i--)
21             d = d - fmod(d, a[i]);
22
23         printf("%.0f\n", d);
24     }
25
26     return 0;
27 }
```

OUTPUT:

TEST CASE 1:

```
Input:
3
3 100
31 71 21
62
4 100
11 16 51 50
44
1 1
1
1
```

TEST CASE 2:

```
Input:
3
5 1000
311 171 211 45 32
622
6 1000
111 161 511 510 56 93
0
2 10
9 5
9
```